

Predicting fine-tuning generalization from activation geometry

Decoding Dan's M -update framing

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The proposal

Represent a *context* as an activation vector v_c (read at the token after the last user-message token) and a *behavior* as an activation vector v_b (read at the token after the assistant-message token). Suppose that, before training, the model's context $\rightarrow$ behavior map is approximately linear,

$$v'_b = M v'_c.$$

Dan's conjecture: after fine-tuning context c' to produce a new behavior v''_b (instead of its original v'_b), the map becomes a rank-one correction of the old one,

$$M_{\text{new}} = M + (v''_b - v'_b) v'^{\top}_c$$

The question is whether something of this shape predicts how training on $c_0 \rightarrow b_0$ generalizes to a different $c_1 \rightarrow b_1$.

What the equation actually claims

The update is a rank-one outer product: the behavior shift $(v''_b - v'_b)$ times the trained context $v'^{\top}_c$. Evaluate the new map at the training point:

$$M_{\text{new}} v'_c = M v'_c + (v''_b - v'_b) \langle v'_c, v'_c \rangle = v'_b + (v''_b - v'_b) \|v'_c\|^2.$$

This exactly hits the target v''_b only when $\|v'_c\| = 1$, so the framing wants to live in *normalized / cosine* space with a fitted scalar gain. Now evaluate at a *new* context c_1 :

$$v_b(c_1) = M v_{c_1} + (v''_b - v'_b) \langle v'_c, v_{c_1} \rangle,$$

so the predicted **change in behavior at any new context** is

$$\Delta v_b(c_1) = (v''_b - v'_b) \cdot \langle v'_c, v_{c_1} \rangle$$

Two falsifiable claims fall out:

1. **Direction is constant.** The behavior shifts in the *same direction everywhere* — the direction it was pushed at the training point. Only the magnitude varies across contexts.
2. **Magnitude $\propto$ context cosine.** How much training on c' bleeds into c_1 scales with the similarity $\langle v'_c, v_{c_1} \rangle$ of their (base-model) context vectors.

The clean bridge to what we already measure

We do not measure v_b directly; we measure $\Delta \log P(\text{marker})$ on-policy at the post-response slot. If the behavior is linearly readable — the persona-vector premise — there is a probe direction w with $\ell(c) = \langle w, v_b(c) \rangle$ for the behavior logit. Then

$$\Delta \ell(c_1) = \langle w, v_b'' - v_b' \rangle \cdot \langle v_c', v_{c_1} \rangle = \underbrace{\langle w, \Delta b \rangle}_{\text{one scalar per behavior}} \cdot \langle v_c', v_{c_1} \rangle.$$

That is a **one-free-parameter prediction**: leakage to every bystander context should be *linear in its base-model cosine to the trained context, with a single shared slope*. Directly testable against the existing leakage tables (#432→#456, #448): regress measured $\Delta \log P(\text{marker})$ per persona against base-model context cosine; check linearity-through-origin with one slope.

This is the mechanism for #404/#458, not a new direction

The persona-distance predictor line (#404/#458: does base-model KL/JS/cosine predict leakage?) is the *empirical* version of exactly this. Dan’s M -update is the generative model that says *why* distance predicts leakage, and it sharpens the existing result three ways:

- predicts the relationship is **linear in cosine**, not merely monotonic;
- predicts a **shared slope** across contexts for a fixed behavior;
- adds genuinely new, untested content — the **directional claim**: the activation-space shift is the *same direction* everywhere. Current predictors only touch magnitude. Whether $\Delta v_b(c_i)$ is rank-one / constant-direction is the part nobody has checked.

It also explains contrastive negatives (#18/#207/#383)

Positive-only training gives $M + \Delta b c_{\text{src}}^\top$. Leakage at bystander c_j is $\Delta b \cdot \langle c_{\text{src}}, c_j \rangle$; since persona contexts share a high baseline cosine, you get *uniform* leakage — exactly the #18/#207 observation (92–98% on all bystanders). Contrastive negatives add **negative rank-one terms** at the bystander contexts that cancel the $\langle c_{\text{src}}, c_j \rangle$ term, which is why the distance→leakage gradient exists *only* in the contrastive regime. The framing predicts this for free:

$$M_{\text{new}} \approx M + \Delta b c_{\text{src}}^\top - \sum_j \varepsilon_j c_{\text{byst},j}^\top.$$

Where it is “a bit wrong” (Dan’s hedge), ranked by impact

1. **Direction rotation is the real risk.** The strong claim is a context-independent shift direction. Plausibly it rotates — evil-on-coding could land as “evil” nearby but “off-target weird” far away. First thing to measure: SVD the matrix of realized $\Delta v_b(c_i)$ and check it is rank-one with right vector $\approx v_c'$ and left vector $\approx \Delta b$.
2. **Rank-one sufficiency.** SGD distributes the update across all weights, so the induced map-change may be higher rank. ROME/MEMIT show narrow edits *are* roughly rank-one, so it is a reasonable prior — but it is the load-bearing assumption.

3. **M itself shifts = emergent misalignment.** The model assumes M is preserved except along v'_c . EM is precisely the failure of that: narrow training broadly rewrites M . In-framing, EM corresponds to either (a) a trained context v'_c with high mean similarity to the whole context distribution (a generic context $\Rightarrow$ broad leakage), or (b) a realized shift with a large off-rank-one component. A concrete, testable account of *when* narrow training generalizes broadly.
4. **Normalization & layer choice.** Needs cosine + fitted gain, and a sweep over which layer / layer-pair gives the cleanest linear M . Context and behavior may read best at different layers.
5. **Lossiness.** A single-token v_c/v_b may not capture the full context/behavior, and emission goes through softmax + sampling. The linear-probe bridge handles the last step, but only if the behavior is genuinely linearly readable.

Extending to the non-linear case

The before-map $v_b = Mv_c$ is a linearization of the true forward map f between the two token positions. Two *distinct* linearizations hide inside Dan’s single M , and separating them is what makes the non-linear extension non-trivial rather than cosmetic.

Input-space Jacobian (describes current behavior). For a smooth f , locally $f(v_c) \approx f(v'_c) + J(v'_c)(v_c - v'_c)$ with $J = \partial v_b / \partial v_c$. So the M in Dan’s *before* equation is really the Jacobian of the frozen model at the operating point. Its predictions hold only *locally*; far from v'_c the second-order term dominates and the shift direction rotates — exactly the “direction-rotation” failure mode, now derived rather than asserted.

Weight-space NTK (describes how training transfers). The quantity Dan actually wants — how training on c' generalizes to c_1 — is not an input-space object at all; it lives in *weight* space. One gradient step on the regression loss $\frac{1}{2}\|f_\theta(c') - b''\|^2$ moves the function at any context c by

$$\Delta f(c) = \eta (b'' - f(c')) \underbrace{\langle \nabla_\theta f(c), \nabla_\theta f(c') \rangle}_{K(c, c')} ,$$

$K(c, c') = \text{empirical NTK}$

and the fully-trained (single-point, min-norm) solution is kernel regression:

$$f_{\text{new}}(c) = f_{\text{old}}(c) + \frac{K(c, c')}{K(c', c')} (b'' - f_{\text{old}}(c'))$$

Dan’s formula is exactly this for a linear model. Take $f(c) = Mc$. Then $\nabla_M f(c) = c$, so $K(c, c') = \langle c, c' \rangle$ and, with $\|c'\| = 1$,

$$f_{\text{new}}(c) = f_{\text{old}}(c) + \langle c', c \rangle (b'' - v'_b),$$

which is Dan’s $\Delta v_b(c) = (v''_b - v'_b) \langle v'_c, v_c \rangle$ verbatim. The dictionary:

- behavior shift $(v''_b - v'_b) \leftrightarrow$ residual $b'' - f(c')$;
- context cosine $\langle v'_c, v_c \rangle \leftrightarrow$ normalized NTK $K(c, c')/K(c', c')$.

So the non-linear extension is: replace the activation cosine with the empirical NTK. The raw cosine is the *linear-model* NTK; the network’s true NTK $K(c, c') = \langle \nabla_{\theta} f(c), \nabla_{\theta} f(c') \rangle$ is the object that actually governs generalization (Malladi et al., *A Kernel-Based View of LM Fine-Tuning*, 2023). For many training points, $f_{\text{new}}(c) = f_{\text{old}}(c) + \sum_i K(c, c_i) \alpha_i$ with α solving the kernel system — the contrastive-negative cancellation falls out as the off-diagonal of the kernel gram, *derived* instead of assumed.

The deep payoff: lazy vs. rich. The kernel picture is exact only in the *lazy* regime, where the NTK stays (approximately) constant through training. If persona / marker fine-tuning is lazy, Dan’s framing-with-NTK is fully predictive. If it is *rich* (feature-learning), training moves the kernel itself, and generalization is *not* fixed-kernel regression — which is precisely the in-framing signature of emergent misalignment (K itself is rewritten; failure mode 3 above). So “can we predict generalization from base-model geometry?” sharpens into the single diagnostic “*is persona fine-tuning in the lazy regime?*” — whose answer decides whether the whole program works.

Two cheaper intermediate extensions.

- **Link function (handles the softmax).** Keep the linear prediction in *logit* space and apply the known output nonlinearity on top: emission rate $= \sigma(\ell_0 + \langle w, \Delta b \rangle \langle v'_c, v_c \rangle)$. A GLM, not a new theory — removes the softmax-lossiness objection for free.
- **Kernelize the activation similarity.** Stay in activation space but replace the raw cosine with $\langle \phi(v'_c), \phi(v_c) \rangle$ for a feature map ϕ (a later / more-processed layer, a learned probe space, attention-pattern features). Let the data pick the similarity that best predicts leakage, without paying for the full NTK.

Testable.

1. Compute the empirical (LoRA-restricted, hence cheap) NTK gram between contexts; check whether $K(c', c_i)$ predicts held-out leakage *better* than the raw cosine. The gap quantifies how much non-linearity matters.
2. **Lazy-vs-rich diagnostic:** measure how much the NTK itself drifts between base and trained model across the persona panel. Large drift $\Rightarrow$ broad EM, and the fixed-kernel prediction should fail exactly where drift is largest.

Caveats. The empirical NTK is expensive in general (a parameter-gradient per context); LoRA makes it tractable (gram over only the adapter parameters). The lazy assumption is the whole ballgame — but its *failure* is informative, not fatal: it localizes where feature-learning (hence broad generalization) actually happens.

Minimal experiment to nail it

One pod, one clean test:

- Fix a behavior (the marker is cleanest — already measured on-policy). Train a single source context $c_0 \rightarrow$ marker.
- Across N held-out contexts (the persona panel), measure: (a) realized $\Delta \log P(\text{marker})$; (b) realized activation shift $\Delta v_b(c_i)$; (c) *base-model* context cosines $\langle v_{c_0}, v_{c_i} \rangle$.

- **Headline test:** predict held-out leakage using *only* base-model quantities plus the single training pair. If that prediction correlates with measured leakage across held-out contexts, the framing is predictive. Report (i) variance explained by the rank-one / cosine model, (ii) whether the shift direction is constant, (iii) the residual = the off-target / direction-rotation part.

Even a partial result (“magnitude is cosine-predictable, direction rotates by X° ”) is publishable and tells us something real.

Take

Worth pursuing, as the *deep* one. It (a) unifies three threads under one generative model, (b) upgrades #404/#458 from “distance correlates” to “here is the mechanism + a directional prediction,” (c) is Dan’s own framing, hence maximally aligned, and (d) is exactly the “fewer experiments, nailed carefully” shape he keeps asking for — feeding the Aug 31 paper-draft milestone. Caveat if it stalls: do not over-invest in the *exact* rank-one form. The valuable deliverable is “to what extent is fine-tuning generalization linearly predictable from base-model activation geometry”; the rank-one update is one hypothesis within that, not the whole prize.